

Module 9

Cost Characteristics and Behavior

Learning Objectives

By the end of this module you should understand:

1. Why cost information is important
2. The distinction between different categories of cost and how these costs behave:
 - Variable and fixed costs
 - Product and period costs
 - Direct and indirect costs
 - Standard and actual costs
 - Controllable and non-controllable costs
 - Engineered and discretionary costs
 - Traceable and common costs
3. The dangers surrounding unitizing fixed costs
4. The construction, and underlying assumptions, of the break-even chart
5. How to calculate the break-even point, and the contribution margin ratio in single-product and multi-product businesses
6. The managerial significance of these techniques

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Review Questions and Case Studies

This module addresses the cost collection process and cost allocation process

Cost Collection



The ingathering of costs

Cost Allocation



The spreading of these costs over the goods and services produced by the organization

The need to determine cost is universal; the difficulties surrounding the determination of cost are considerable

Why is cost information so important?

With a knowledge of cost, managers can:

- 1 **Control** actual performance against planned performance and take corrective action
- 2 **Plan** next year's costs carefully, making due allowance for inefficiencies and unforeseen events which distorted last year's performance
- 3 Determine a desirable selling **price** (whether in terms of ticket price in the market or subsidy sought from local or central government), even though that price may not be achievable
- 4 **Track** the consumption of the organization's resources to ensure that all employees are carrying out their duties efficiently and honestly
- 5 **Choose** among alternative courses of action (while recognizing, of course, that decisions are made on many more criteria than cost)

Cost is one of the most fundamental control mechanisms in a management information system

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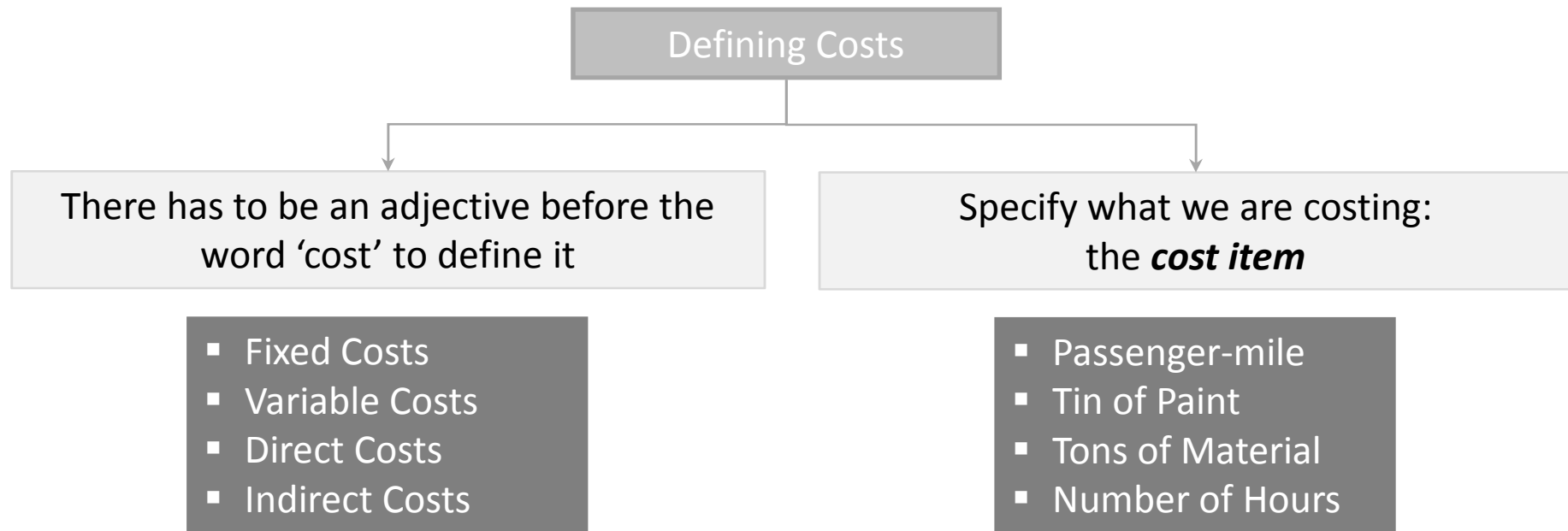
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Review Questions and Case Studies

To understand how costs move we have to first define what our costs are



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We will use last module's example of the DVD cabinets company in our study for costs in this module

Example for Discussion

- The cost of one shelved DVD unit would be stated thus:

One timber-shelved DVD unit

Direct material	€
Chipboard	10.00
Veneer	3.00
Metal trim	2.00
	<u>15.00</u>
Direct labour 1.5 hours @€5 per hour	7.50
Prime cost	22.50
Manufacturing overhead	2.50
Manufacturing cost	<u>25.00</u>
Share of non-manufacturing overhead	5.00
Total cost	<u><u>30.00</u></u>

To understand what fixed and variable costs are, we have to define at first the overhead costs; that's in light of last module's example of the DVD company

Overhead

Costs incurred by a business in support of the cost items being made but not easily identified in the product as materials and labor

Manufacturing Overhead

- Depreciation on saws, lathes, turners and power screwdrivers
- Lubricants and energy costs for the above equipment
- Rent and rates and other space costs associated with the workshop
- Salaries of foremen, supervisors, quality inspectors and production management team

Non-Manufacturing Overhead

- Depreciation on office equipment, computers and motor vehicles
- Rent, rates and other space costs associated with office and showroom
- Salaries of office staff and general management

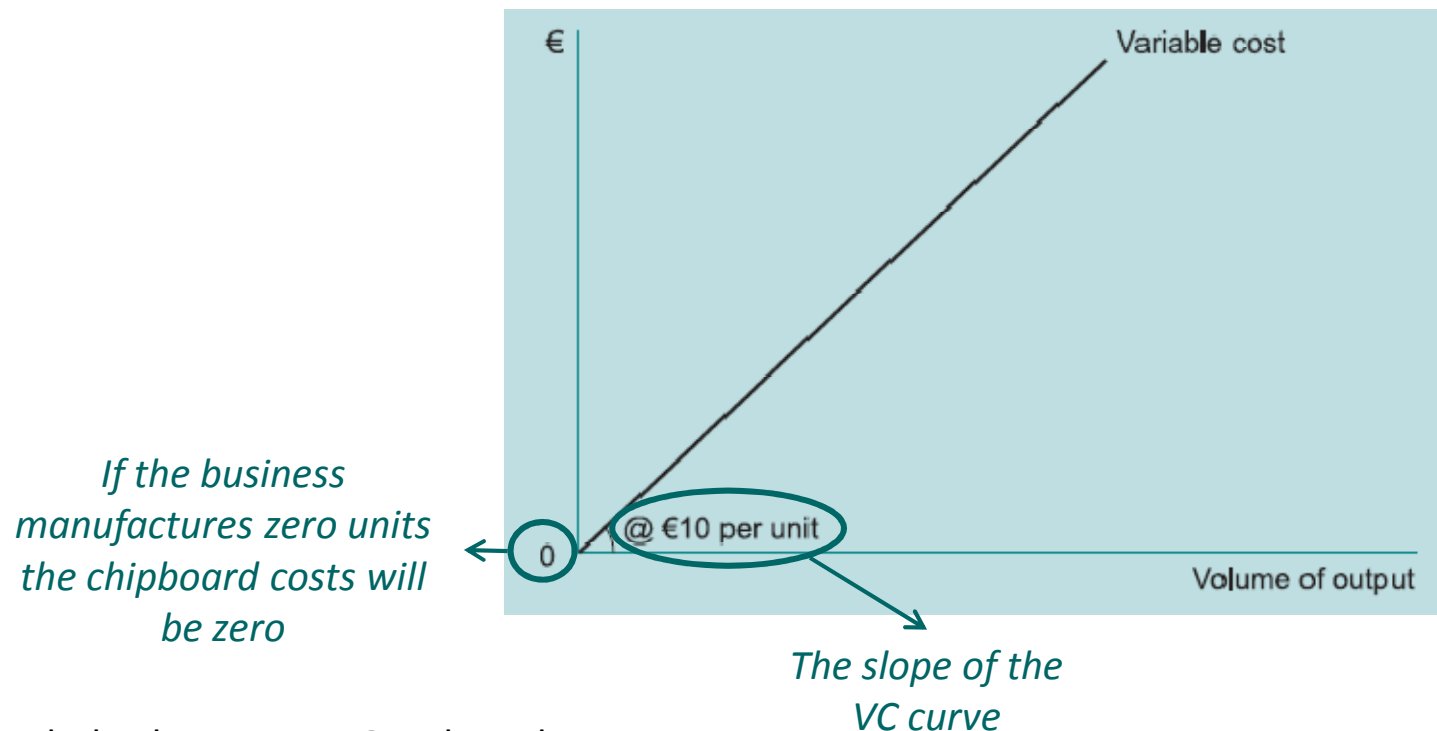
A close examination of all of these costs indicates that some increase with production levels while others do not

Variable costs are the costs that vary directly with volume produced

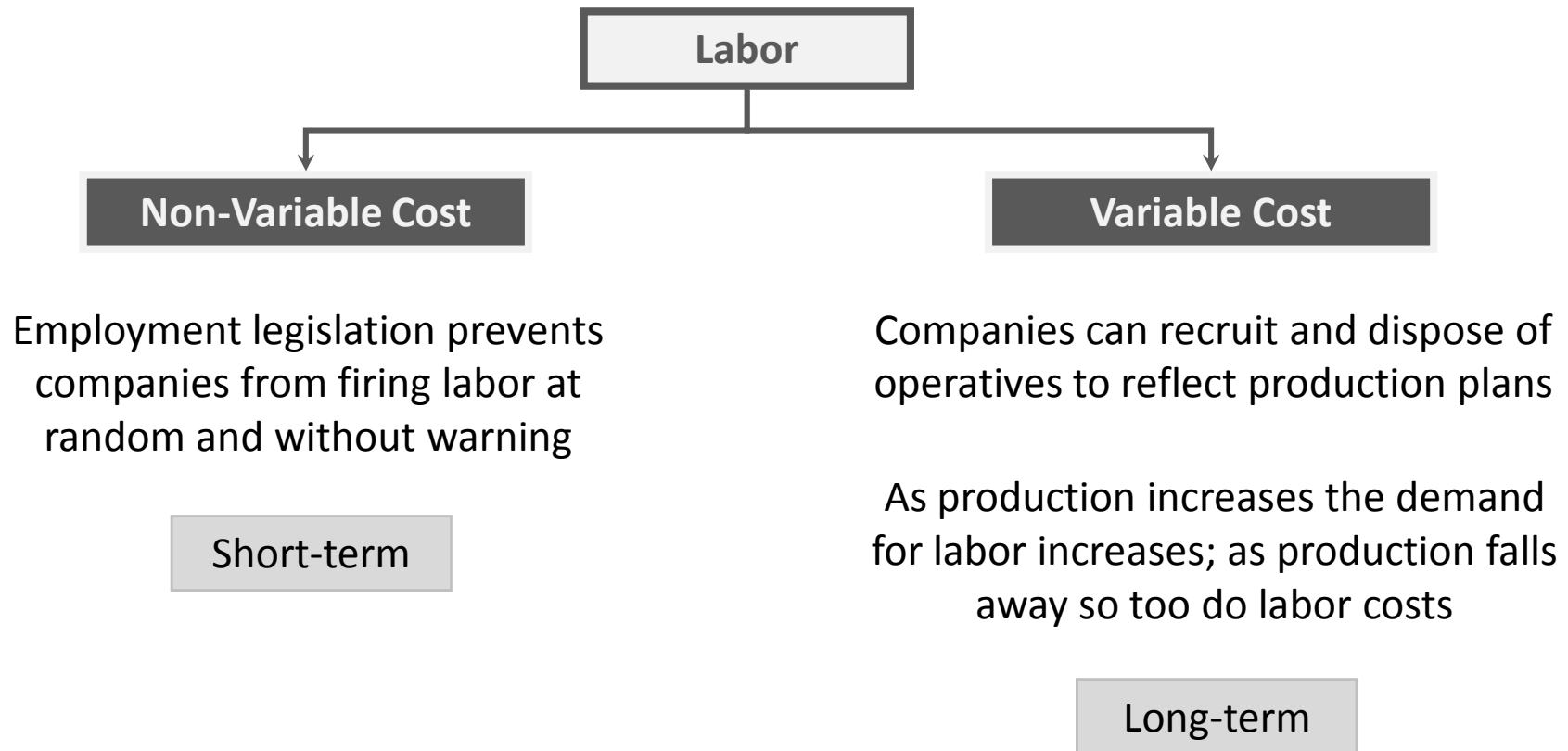
Example

Take chipboard:

- Each unit may require two square meters of timber at a cost of €5 per square meter
- Therefore total cost of €10 per cost item
- If the firm manufactures 10 units the material cost for chipboard will be €100; if 20 units then €200



There are plenty of other variable costs in the DVD unit like veneer, metal trim and labor; labor costs need explanation in this context as it can be treated in 2 ways

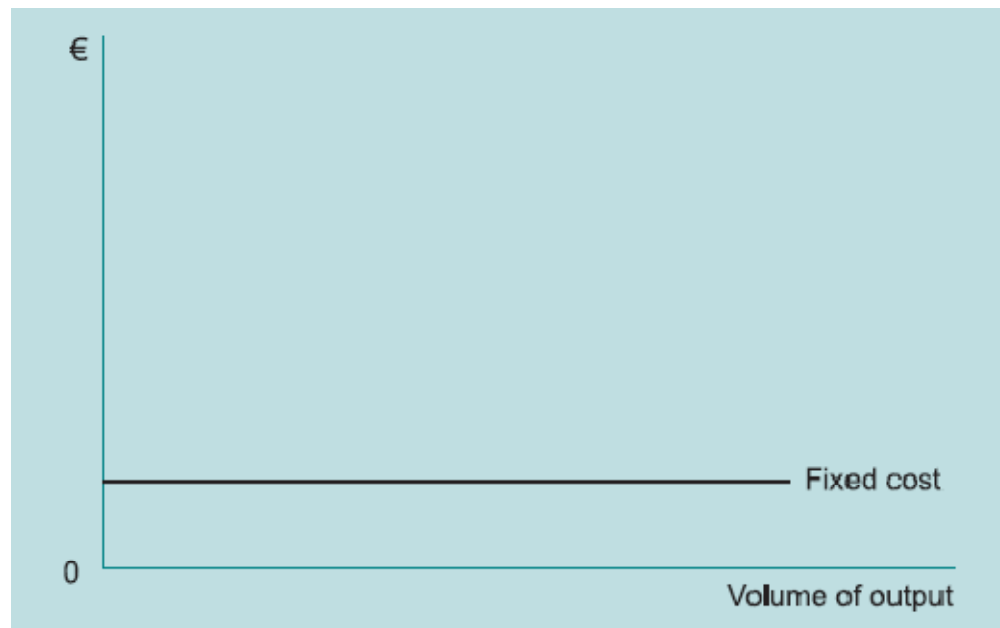


Fixed costs are the costs that are indifferent to the production level

Example

Managerial salaries:

- Such costs are paid regardless of volume level
- Even if production fell to zero (e.g. during a strike) managers would still expect to receive their salaries
- At the other end of the production scale, the same salary would be paid when output reached maximum capacity



Depreciation is treated as a fixed cost although there are some people that argue that it should not be

Variable Cost Argument vs. Fixed Cost Argument:

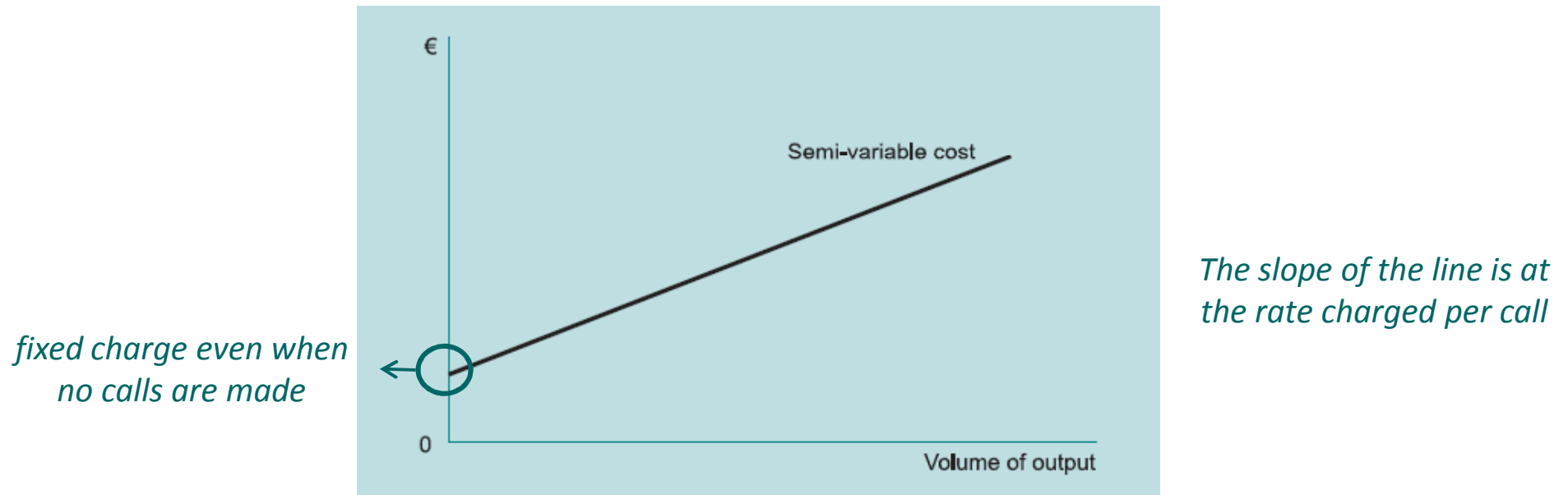
Variable/ Fixed	Argument	But	?
Variable Cost	Machinery depreciates more quickly if the usage is greater	Scenario of zero production rarely happens	x
Fixed Cost	Machines are used on a relatively predictable pattern	Depreciation does not vary with production line	✓
	What causes depreciation is the passage of time that brings it to obsolescence		

Some costs such as the telephone or some energy supplies or certain suppliers of photocopying facilities have features of both variable and fixed costs

Example

The telephone:

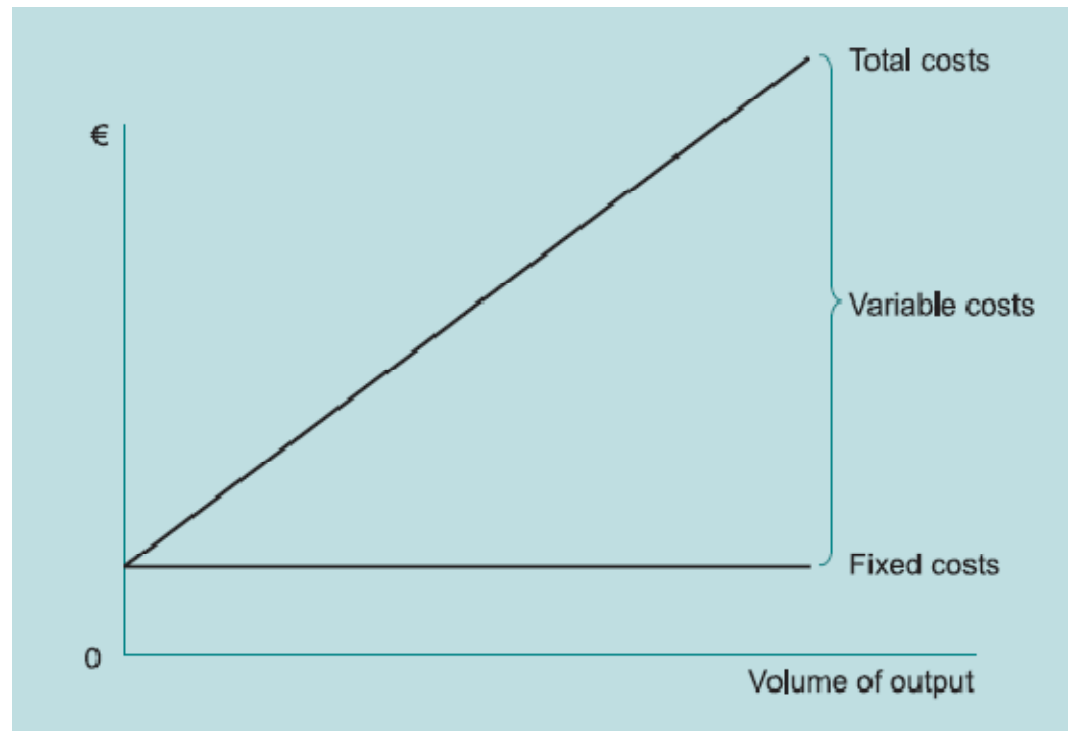
- A fixed charge per annum for the use of the service
- Plus a variable charge that depends on the number and duration of calls made



Accountants tend to disregard semi-variable costs and work with only the two principal categories because of the low number of costs involved

Managers can therefore read the total costs involved at any given production level from the total costs chart

The total costs for a period of time



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Review Questions and Case Studies

When total variable costs and total fixed costs are split into individual products; variable costs seem fixed while the fixed costs vary

Example

The management of a business is considering the level of production to select for the next quarter-year. Sales are buoyant and any of the production levels set out below can be sold

Alternative levels	100	200	300
Costs in total	€	€	€
Variable	100	200	300
Fixed	600	600	600

When management calls for the unit costs the following numbers are presented:

Alternative levels	100	200	300
Costs per unit	€	€	€
Variable	1	1	1
Fixed	6	3	2

Each unit of product contains the same amount of materials and labor regardless of production level, while the fixed burden of overhead is shared out more and more as the production level increases

In practice it is not always obvious when fixed costs have been ‘unitized’ –in terms of when they do stop varying

Example (Continued)

Suppose management selects the upper level of 300 for the next quarter

€1 variable cost
+ €2 fixed cost

€3 Unit cost per item

↓
300 units x €3

€900

Suppose that the business wants to push up production to 400 for the succeeding quarter-year

€1 variable cost
+ €1.5 fixed cost

€2.5 Unit cost per item

↓
400 units x €2.5

€1,000

Some managers would multiply 400 by €3 to determine total cost for the next quarter which is wrong as fixed costs of €600 would be spread even over 400 units (€1.50 per unit)

Managers have to fully understand the concept of unitizing fixed costs, the following case study shows why

Case

You have been approached by a friend who works for a local company manufacturing metal housing boxes for heavy duty electric meters. Over the past two months he has been in charge of a department that makes four styles of casing on one assembly line for use by another department. He was appointed acting manager while the full-time manager recovered from illness. The chief executive, in giving him his first managerial responsibility, charged him with 'keeping costs to the lowest possible level'.

He tells you that shortly after taking over responsibility he was presented with a product cost statement revealing the numbers below. Coincidentally he received a competitor's price list which revealed that product X would be cheaper to buy in.

Product	W	X	Y	Z
Costs per unit	€	€	€	€
Materials + labour	7	11	12	6
Manufacturing overhead	4	3	2	1
	11	14	14	7
Competitor's price	12	12	16	9

The manager has made the mistake of believing that when Product X dropped out all of X's costs would drop out too

Anxious to follow his chief executive's instructions, and thereby to make a good impression, your friend signed a 12 months' contract for Product X with the competitor and closed down the X line immediately. 'As you can see,' he explains to you, 'I have saved the company €2 for every casing ordered.'

But he was concerned with the latest print-out of costs received a few days later. Product W now looked expensive to make against the competitor's prices which had not moved. 'The problem is that the manager of the department is returning to work next week and I'm slightly concerned that if I get rid of W I'll show up my colleague to be naïve in trying to make all four casings. I don't want to do this but I would like to save the company even more money.'

Product	W	X	Y	Z
Costs per unit	€	€	€	€
Materials + labour	7.00		12.00	6.00
Manufacturing overhead	5.71		2.86	1.43
	<u>12.71</u>		<u>14.86</u>	<u>7.43</u>
Competitor's price	12	12	16	9

Instead of saving the company €2 per product, it cost the company €1 (the difference between the price paid to the competitor and the variable costs of manufacturing)

Further investigation will enable you to show your friend the error he made

Further Investigation

You determine that manufacturing overhead (assumed all fixed costs) amounts to €1,000 for each month. Overhead is spread over products using machine hours. When your friend assumed command, 1,000 machine hours were used per month, i.e., €1 manufacturing overhead per machine hour

All products

	W	X	Y	Z	Total
No. of products made	100	100	100	100	
No. of machine hours	4	3	2	1	
Overhead allocated	€400	€300	€200	€100	€1000

Dropping X

	W	X	Y	Z	Total
No. of products made	100		100	100	
No. of machine hours	4		2	1	
Overhead allocated	€571.42		€285.71	€142.87	€1000
Overhead per unit	€5.71		€2.86	€1.43	

If he drops Product W he would be compounding his earlier mistake and it would not be long before the entire department would be closed

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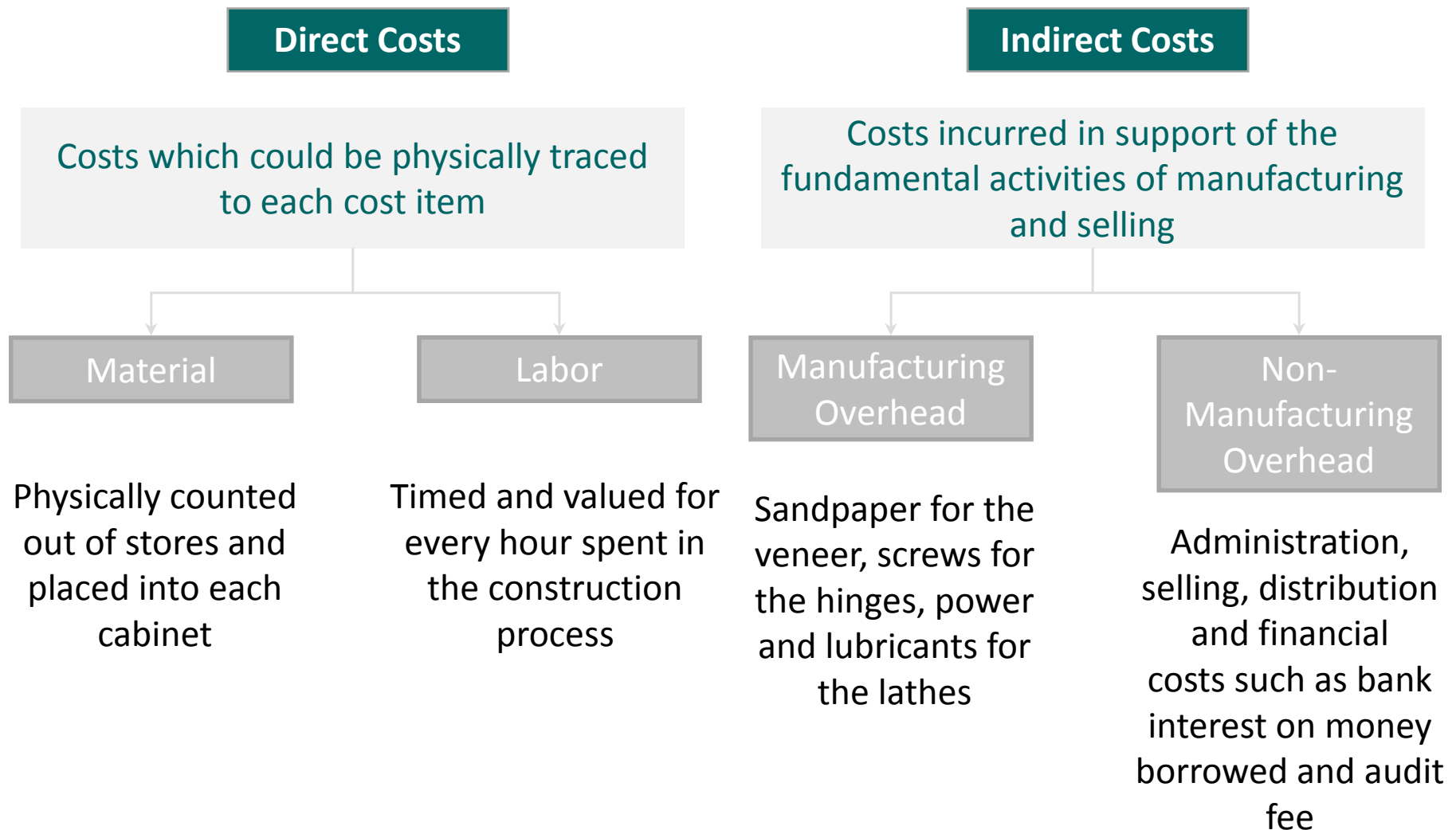
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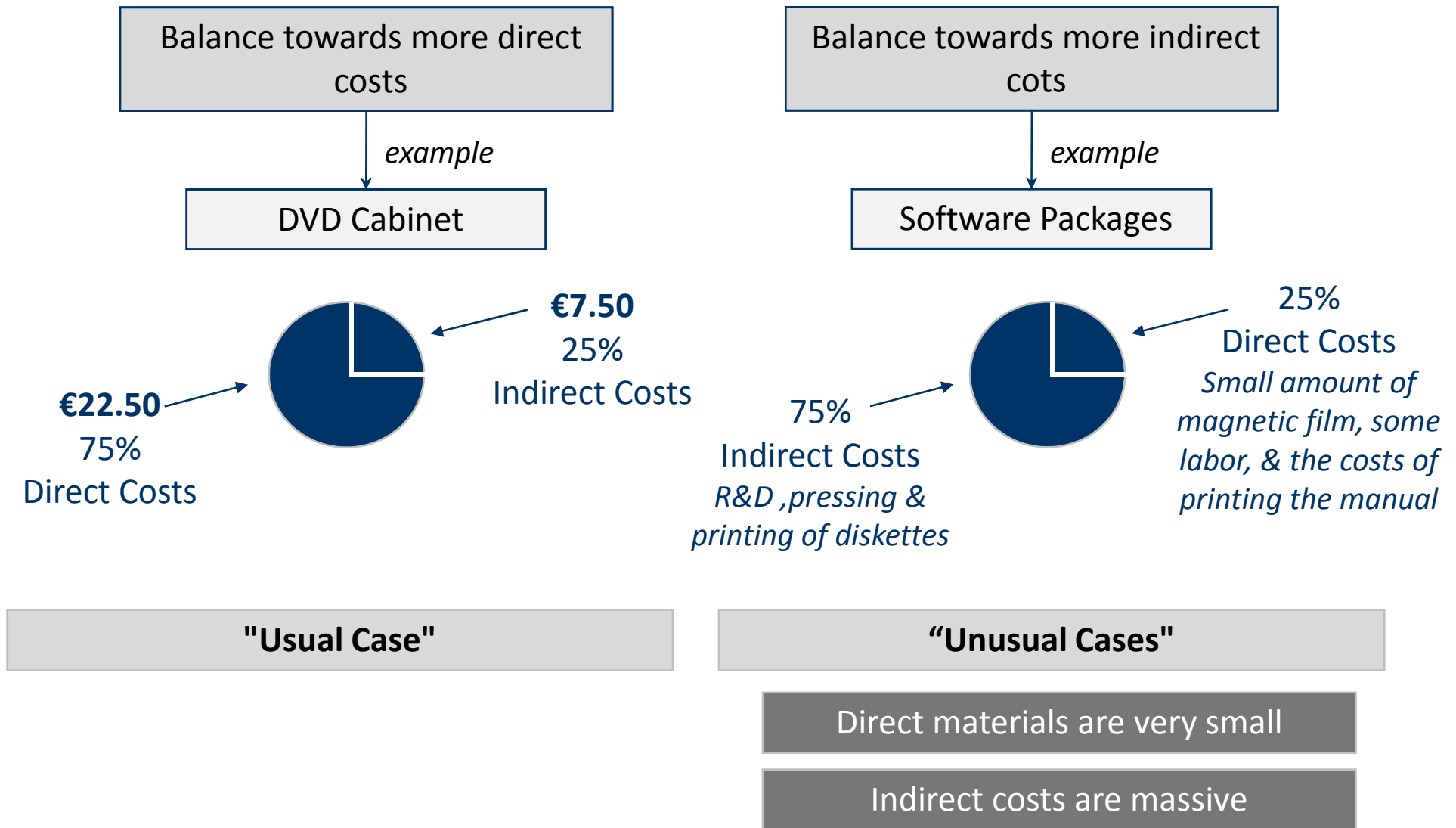
Review Questions and Case Studies

Another way of separating costs is to consider their relationship with the cost item –direct and indirect costs

Consider the cost profile of the timber-shelved DVD cabinet

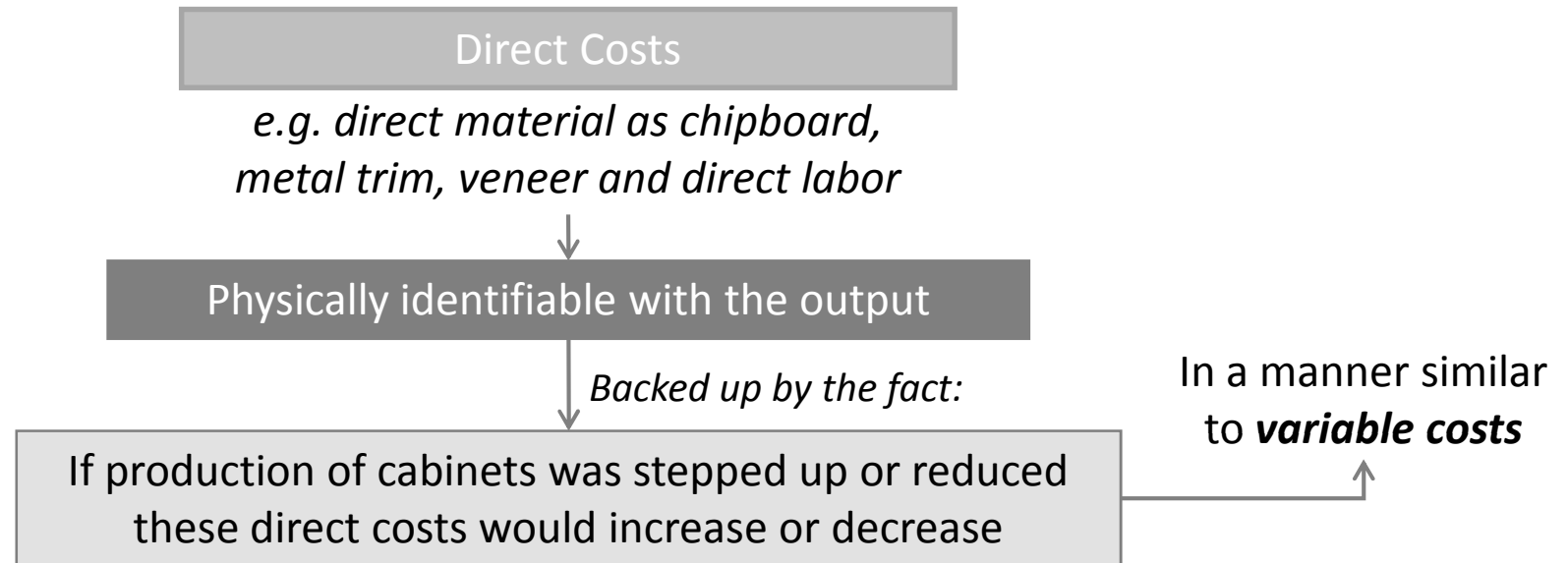


There are some cases where the balance of total costs swings more towards indirect costs



Are direct and indirect costs the same as variable and fixed costs? The answer is, not necessarily, but there is a significant measure of overlap

In the DVD cabinet example



Therefore a close correspondence between direct and variable costs

↓ *But*

Sometimes this closeness is broken by a direct cost that is a fixed one

Example

The DVD cabinet factory produced a number of lines of foldaway flatpack furniture. It required a special tool to mount the castors. This tool is a fixed cost (in terms of depreciation) but is a cost directly attributable to the production of DVD cabinets

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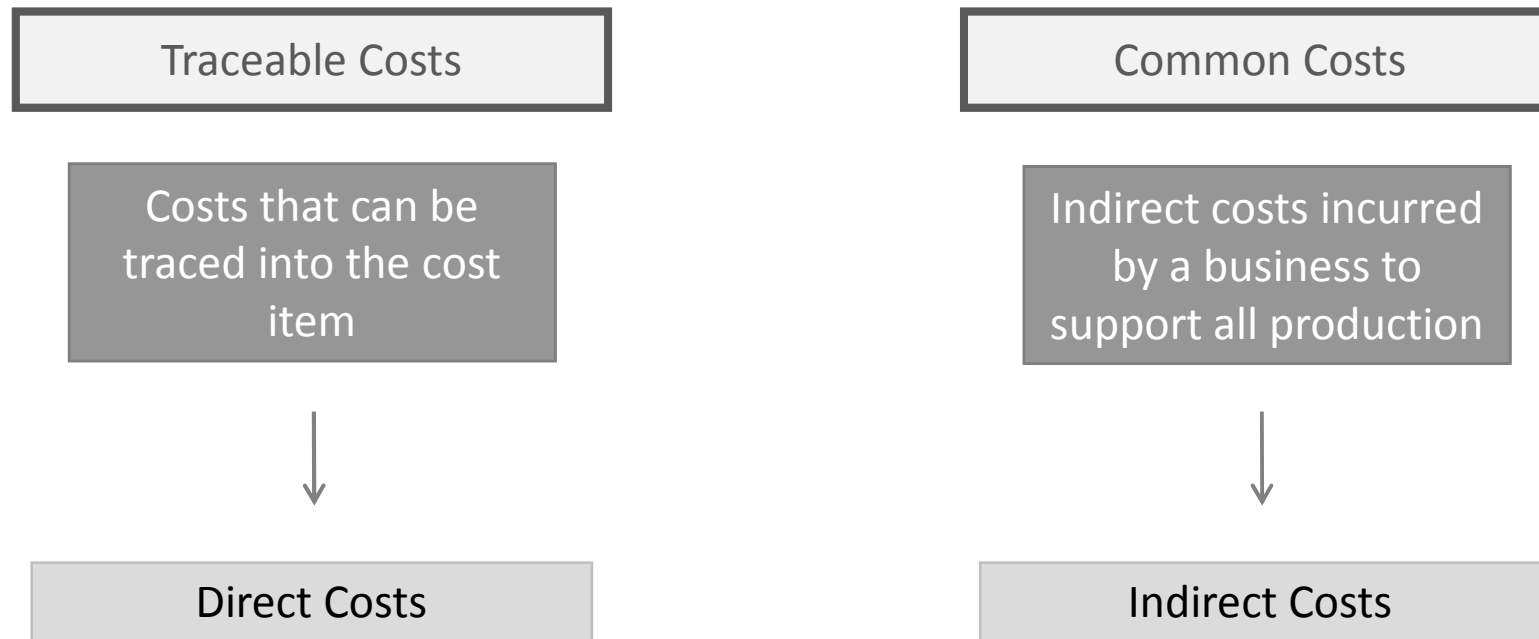
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A more descriptive, but less widely used, category of costs is traceable and common costs



if a business produces only one product line, it will not have any common costs because every cost can be directly traced to the cost items produced

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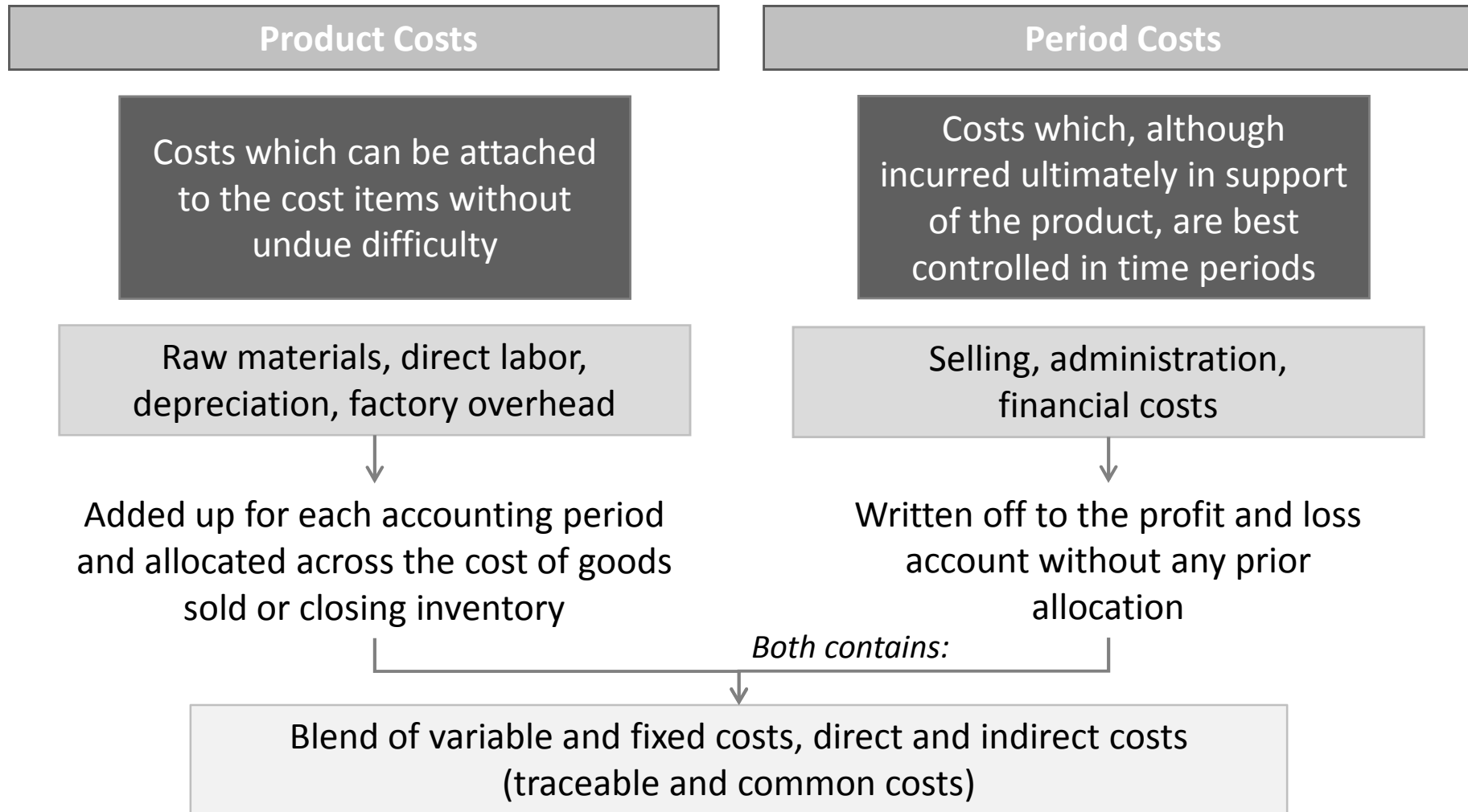
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Costs can also be categorized into product and period costs



This cost category helps value inventory for financial reporting purposes. It is not a useful tool for managerial planning and control

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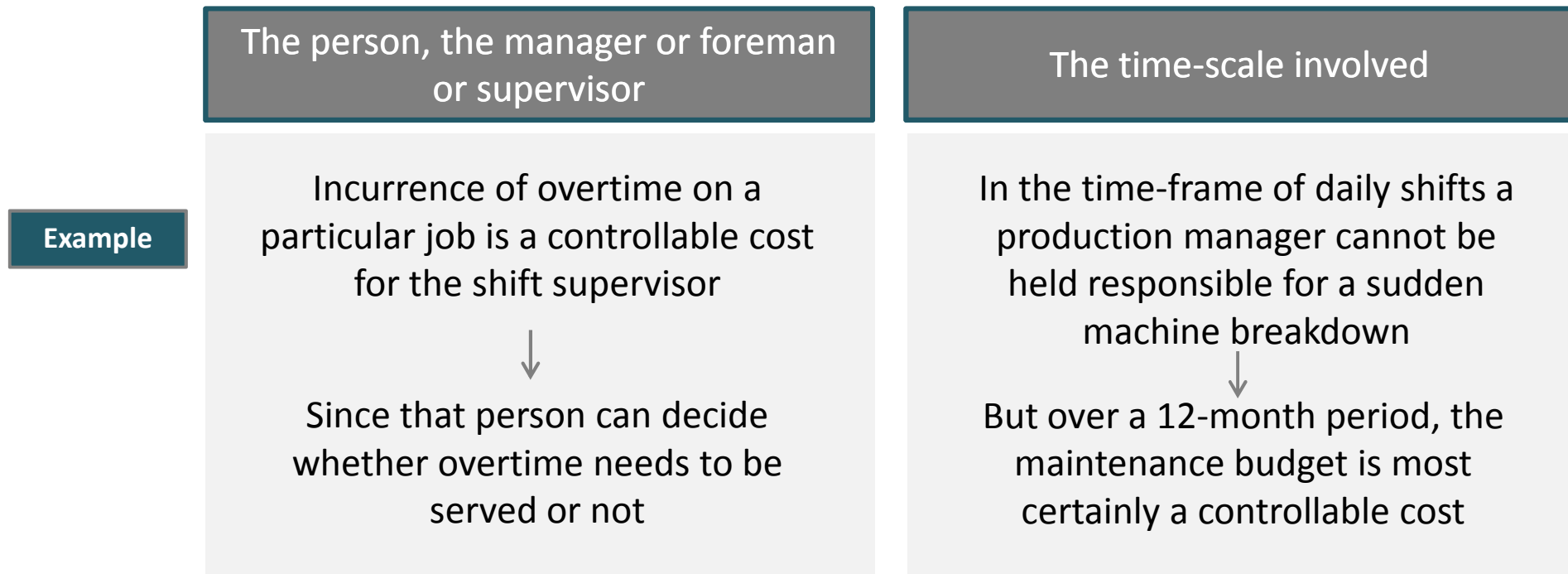
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Review Questions and Case Studies

In this cost category, the focus moves away from the cost item, e.g. the DVD cabinet, and moves to the individuals in an organization who incur and control costs

The concept of control is influenced by:



It is sometimes said, 'In the short term virtually no cost is controllable by anyone!'

Exercise

Is the insurance premium on the factory buildings and plants of the factory a controllable cost or not?

The person, the manager or foreman or supervisor

Controllable by a senior manager in a corporate position such as the chief accountant or financial director

The time-scale involved

Even over a period of a year the financial director cannot control such an expense



Considerable planning and advice would be needed before a reduction in this type of cost could be effected

Hence it is as said, 'In the short term virtually no cost is controllable by anyone!'

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Review Questions and Case Studies

Before a business can assess its performance, it must know the benchmarks against which its achievements can be measured –which is what we call standard costs

Standard Costs

Budgeted costs for one cost item

Detailed costs for individual products and services set up by a business to act as a benchmark

Based not only on engineering studies but on recent experience

Actual Costs

Actual Costs incurred



Against this norm, the actual costs are measured period by period

Management would want to receive explanations for significant variances between the two sets of figures

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Review Questions and Case Studies

Discretionary costs are those costs that can be influenced by managerial decision

Engineered Costs

Costs engineered into the product

Should be incurred every accounting period

e.g.: the cost of chipboard veneer, metal trim and direct labor or certain critical items of manufacturing overhead

Discretionary Costs

Costs that managers can reduce relatively quickly

Need not be incurred every accounting period

e.g.: machine maintenance, administrative support and R&D

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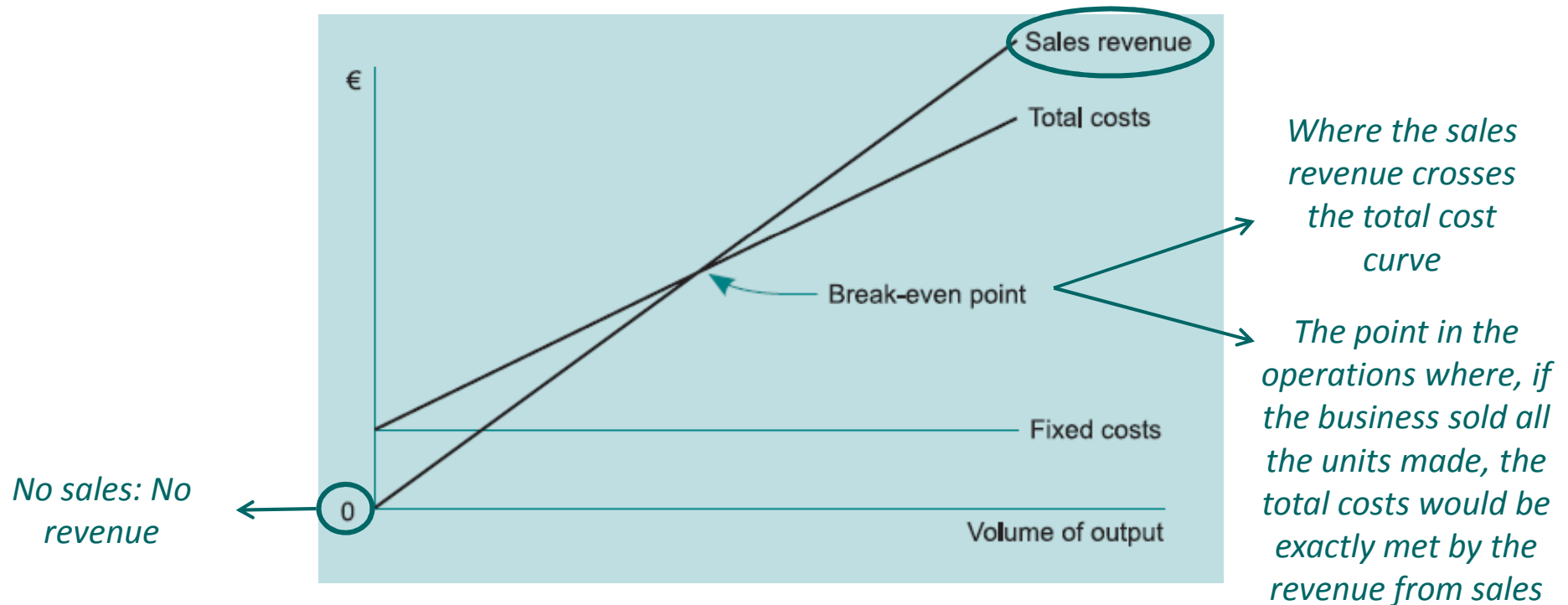
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To draw a break-even chart we add the sales revenue line to the fixed and total cost curves

The sales revenue line climbs from zero at the rate of the revenue earned for each unit of output sold



The break-even chart develops the relationships between cost and output; we call it the cost-volume-profit analysis

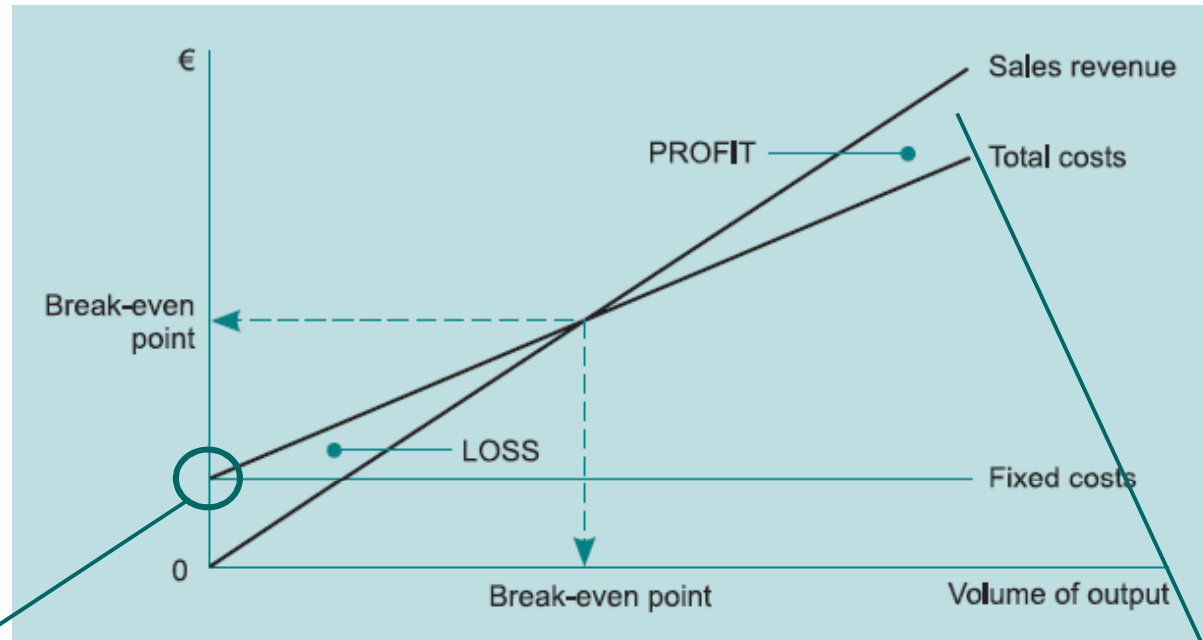
At the break-even point the business makes neither a profit nor a loss

Loss

If a company fails to operate at its break-even point



The total costs exceed sales revenue and a **loss** is incurred



Maximum loss will be incurred when the company has zero sales and the total costs equal the fixed costs

Profit

If the company pushes production beyond break-even point



Profit increases as the sales revenue line pulls away from the total costs

Margin of Safety

The difference between the break-even point and the actual level of output achieved, when greater than break-even

The break-even point can be read off the chart in terms of either units or money

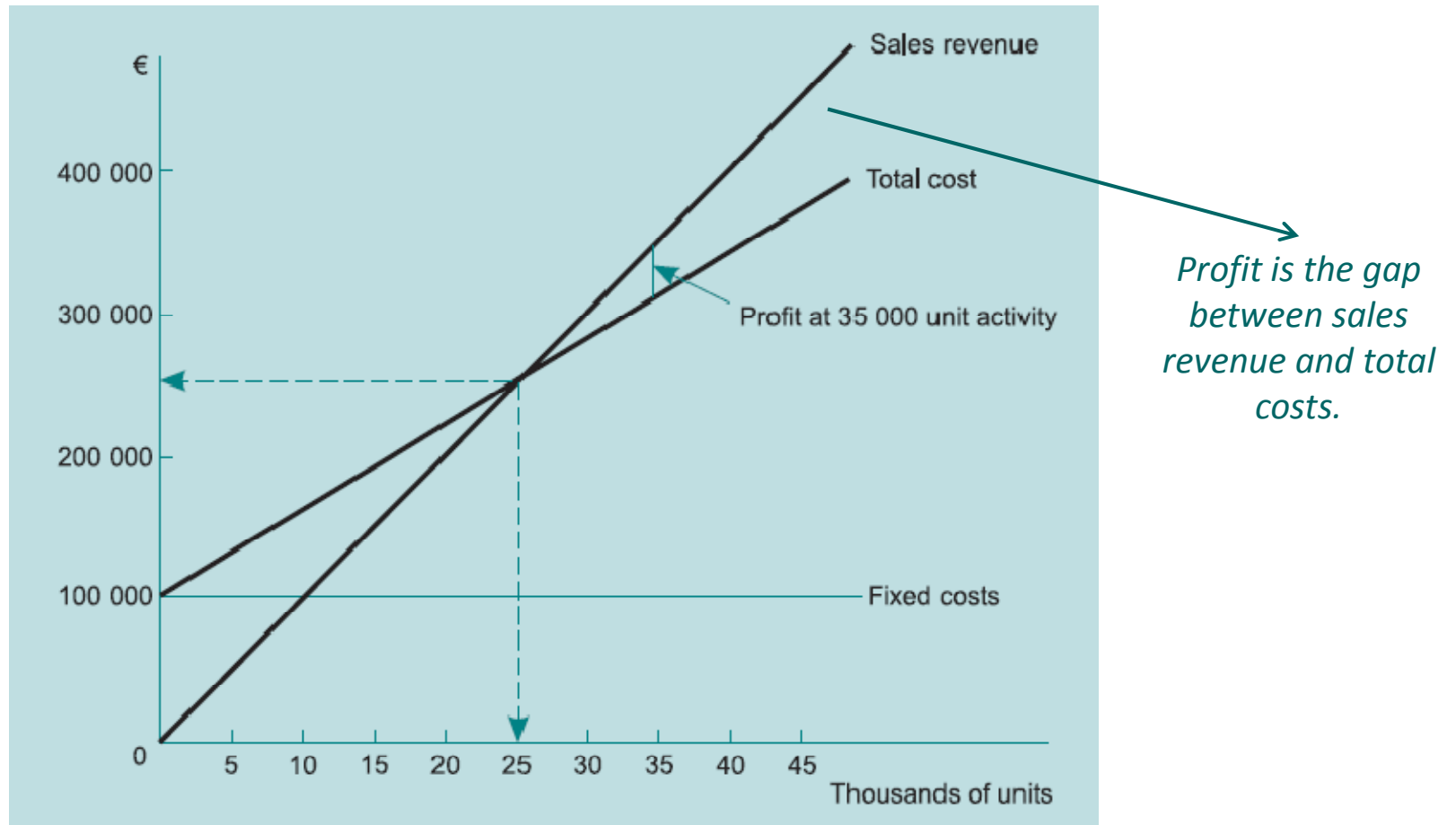
Example showing how to draw a break-even chart

Example

A business manufactures and sells a component for use in the assembly of disk drives. The selling price is €10 per unit; variable costs are €6 per unit and fixed costs are €100,000 per year. What is the break-even point in terms of volume of units sold and money received?

1. Plot the fixed costs curve at the €100,000 level
2. Plot the total cost curve using two sample levels of operations:
 - (a) at 10,000 units, total costs would be $€100,000 + (10,000 \times €6) = €160,000$;
 - (b) at 30,000 units, total costs would be $€100,000 + (30,000 \times €6) = €280,000$.
3. Plot the sales revenue curve using two sample levels of operations:
 - (a) at 10,000 units, revenue would be $(10,000 \times €10) = €100,000$;
 - (b) at 30,000 units, revenue would be $(30,000 \times €10) = €300,000$.

The break-even point can be read off the graph as 25,000 units or €250,000 sales revenue



What profit would be made if the business made and sold 35,000 components?

At the 35,000 output level we read from the chart that revenue would be €350,000 and costs would be €310,000, yielding a profit of €40,000.

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Managers have a choice between the cost structures within which they may operate; this affects the break-even point and inspires managers to take greater or lesser risk

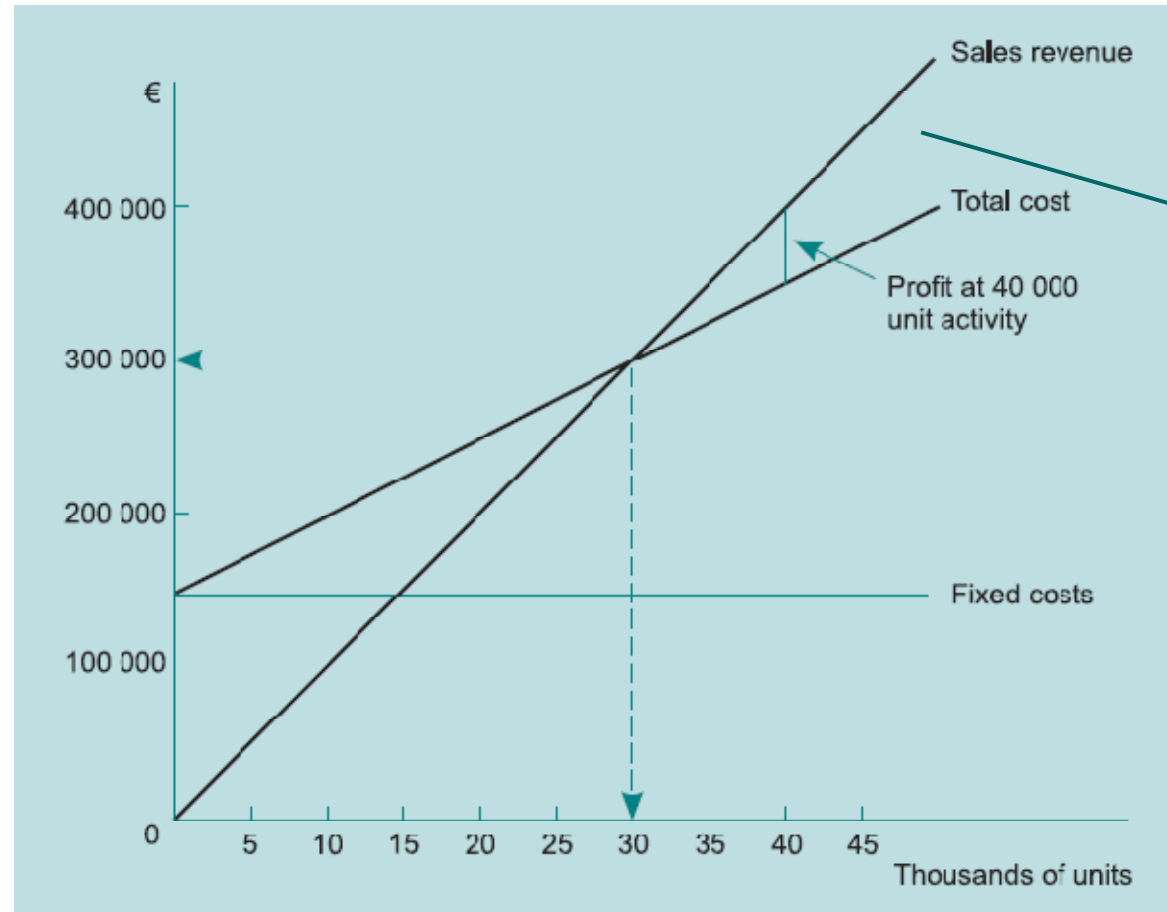
Example

Assume that for next year the management of this business is faced with the possibility of investing in substantial new production equipment which would have the effect of driving up fixed costs (i.e. depreciation and maintenance) by €50,000 but would also reduce variable costs by €1 to €5 per unit (i.e. less skilled labor required in final assembly).

Consider the break-even chart for this cost profile

1. Plot the fixed cost curve at the €150,000 level.
2. Plot the total cost curve using two sample levels of operations:
 - (a) at 10,000 units, total costs would be $€150,000 + (10,000 \times €5) = €200,000$;
 - (b) at 30,000 units, total costs would be $€150,000 + (30,000 \times €5) = €300,000$.
3. Plot the sales revenue curve as before.

The break-even point moved up from 25,000 units to 30,000 but if the business pushes up volume by another 10 000 the profit is €50,000 (sales € 400,000 less total costs € 350 ,000)



The rate at which they earn profit beyond the break-even point is quicker

Managers who are confident of selling well beyond their break-even volume, or are willing to take a risk, would select the second cost structure

This can be quantified by using the profit/volume ratio which measures the impact of volume on profit

$$\text{P/V ratio} = \frac{\text{Profit}}{\text{Sales price}} = \frac{\text{Sales price} - \text{Variable costs}}{\text{Sales price}}$$

Cost Structure 1

$$\begin{aligned} \text{P/V} &= \frac{€4}{€10} \\ &= 0.4 \end{aligned}$$



In the start-up phase of an operation when the sales forecasts are relatively uncertain, cautious managers would normally select this because they would reach break-even point 5,000 units earlier

Cost Structure 2

$$\begin{aligned} \text{P/V} &= \frac{€5}{€10} \\ &= 0.5 \end{aligned}$$



Products under this cost structure earn a higher contribution

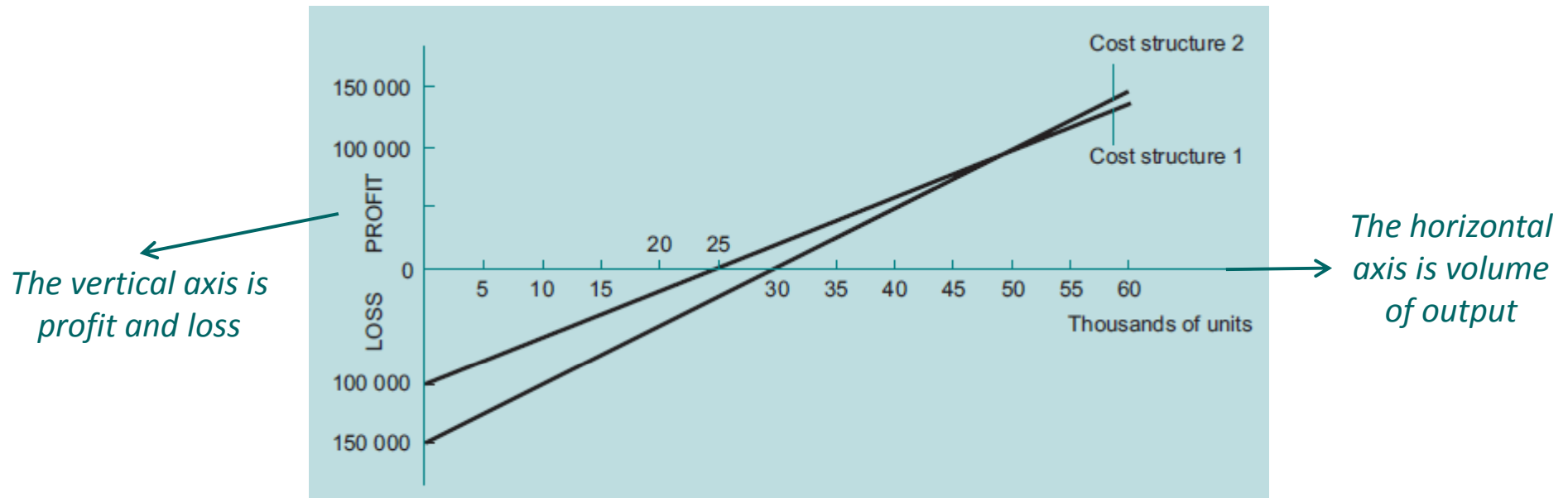
A P/V ratio of 0.4 indicates that for every €1 of sales, products under Cost Structure 1 earn a contribution of €0.40

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Another method of representing costs, volume of output and profit (loss) earned graphically is to feature only profit and volume without costs



The profit line climbs at the rate at which the business earns contribution on each unit (sales revenue per item less variable costs per item)

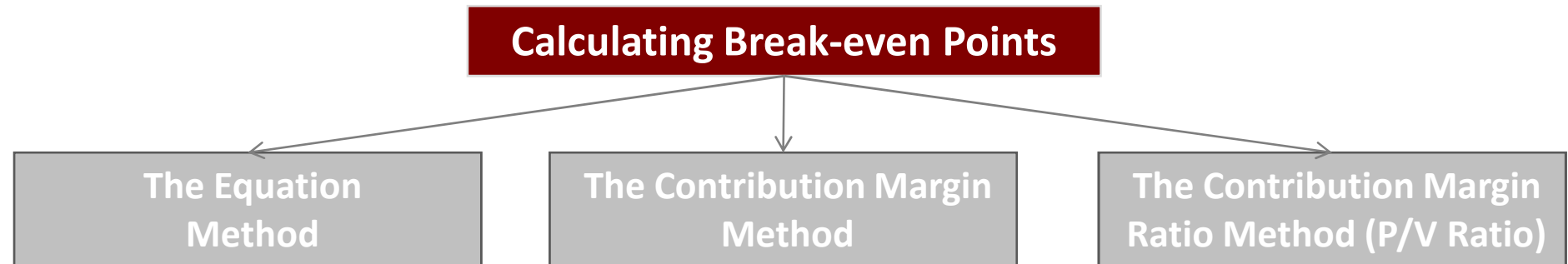
When the profit line cuts the horizontal axis the business is operating at break-even

Agenda

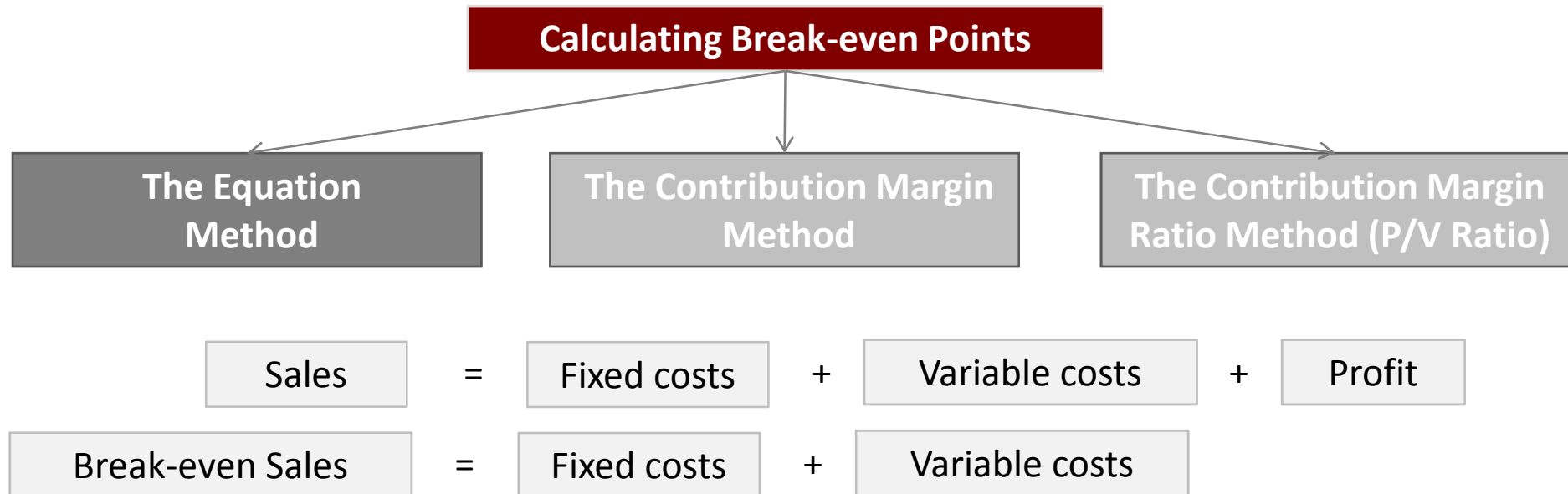
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Review Questions and Case Studies

There are three ways to calculate break-even points other than graphical representations



There are three ways to calculate break-even points other than graphical representations



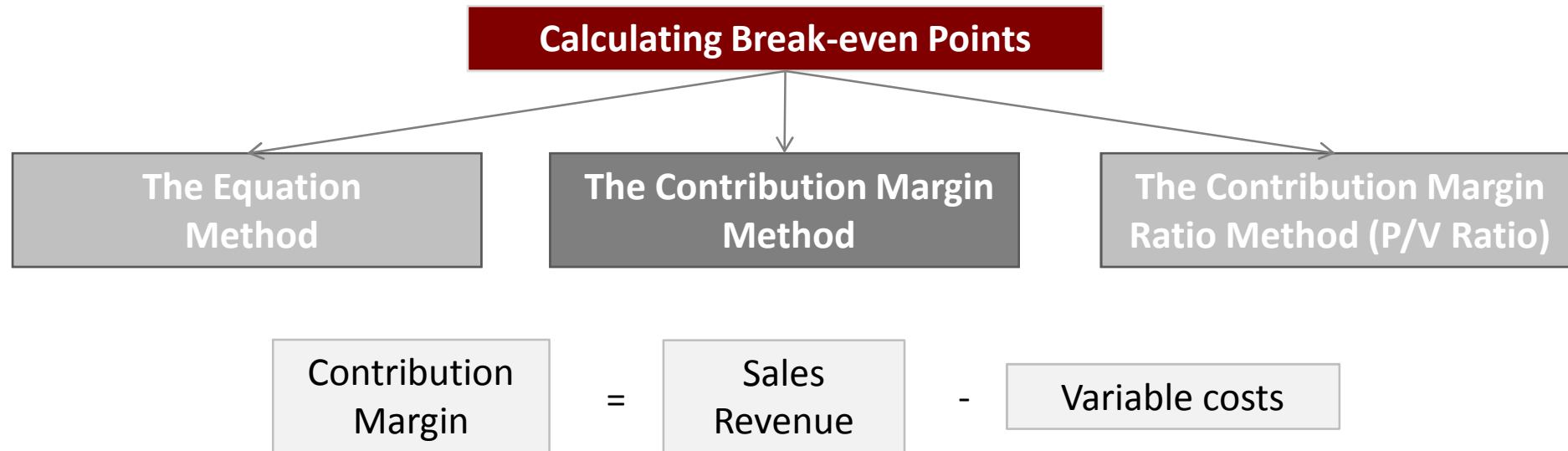
In our example above:

$$S = €100\,000 + 0.6S$$

$$0.4S = €100\,000$$

$$S = \frac{€100\,000}{0.4}$$
$$= €250\,000$$

There are three ways to calculate break-even points other than graphical representations



In the example above sales revenue was €10, variable costs €6 and contribution €4. If each unit earns €4 contribution (towards fixed costs) how many units must be sold to reach the break-even point (BEP)?

$$\begin{aligned}\text{BEP} &= \frac{\text{Fixed costs}}{\text{Contribution margin}} \\ &= \frac{€100\,000}{€4} \\ &= 25\,000 \text{ units}\end{aligned}$$

Example showing how to apply the contribution margin method

Example

In the above example (together with the amended cost structure) how many units must be sold if management wishes to make €200,000 profit?

(Hint: treat the target profit as being similar in nature to fixed costs.)

$$\frac{€100\,000 + €200\,000}{€4}$$

Cost structure 1

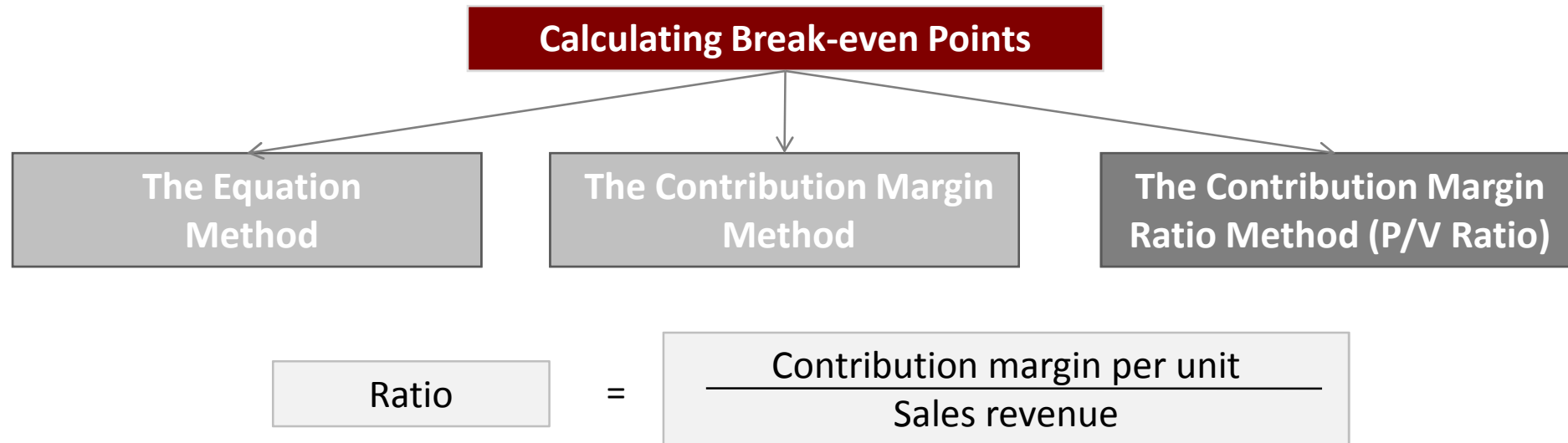
$$\begin{aligned}\text{Sales required} &= \frac{\text{Fixed costs} + \text{Target profit}}{\text{Contribution margin}} \\ &= \frac{(€100\,000 + €200\,000)}{€4} \\ &= 75\,000 \text{ units}\end{aligned}$$

Cost structure 2

$$\begin{aligned}\text{Sales required} &= \frac{\text{Fixed costs} + \text{Target profit}}{\text{Contribution margin}} \\ &= \frac{(€150\,000 + €200\,000)}{€5} \\ &= 70\,000 \text{ units}\end{aligned}$$

Note the impact of a higher fixed-cost threshold combined with a larger contribution margin – Cost Structure 2 gets there more quickly!

There are three ways to calculate break-even points other than graphical representations



For the two cost structures so far dealt with we calculated the ratio to be 0.4 and 0.5 respectively. This ratio can be used in an earlier identity to calculate the break-even point in sales revenue, that is, if each unit has a contribution margin ratio of 0.4 how many sales must be achieved to break even?

$$\begin{aligned} \text{BEP} &= \frac{\text{Fixed costs}}{\text{Contribution margin ratio}} \\ &= \frac{€100\,000}{0.4} \\ &= €250\,000 \end{aligned}$$

Case on Break-Even Analysis

An oil company is considering the siting of a new petrol filling station beside a city bypass. It has made the following calculations:

Cost of operations per annum:	€
Salaries of staff	30 000
Depreciation of equipment	5 000
Rental of site	5 000
Cost of litre of petrol	0.48
Selling price per litre	0.50

1. How many litres must be sold to break even?
2. If an investment in promotional advertising would produce an increase in litres sold of 250 000 litres, how much could the company afford to spend on this exercise to break even?
3. Were the company to sell higher-grade petrol, retailing at €0.52 per litre and costing €0.49 per litre, the more expensive pumps would require a further depreciation of €3000. What effect would such a scenario have on the breakeven point calculated in Requirement 1?

Case on Break-Even Analysis

1. All the operational costs are fixed costs apart from the cost per litre. Therefore the black cloud of fixed costs hanging over the petrol station is €40,000 per annum. Each litre makes a contribution of €0.02.

$$\begin{aligned}\text{BEP} &= \frac{\text{Fixed costs}}{\text{Contribution margin}} \\ &= \frac{€40\,000}{€0.02} \\ &= 2\,000\,000 \text{ litres}\end{aligned}$$

This figure gives management its first yardstick against which to assess the feasibility of this potential site. Market research and previous experience will indicate whether or not this is an achievable figure. If not then the company will be forced to look for a cheaper site or put money into an effective advertising campaign (see Requirement 2).

Case on Break-Even Analysis

2. An extra 250,000 litres sold would yield €5000 ($250,000 \times €0.02$) additional contribution. Therefore the company could afford to spend €5000 on a promotional campaign.

3.

$$\text{BEP} = \frac{€43\,000}{€0.03}$$

$$= 1\,433\,334 \text{ litres}$$

This figure may prove to be a more realistic target for a new filling station but of course it comprises higher-grade petrol which may, by itself, dampen demand.

Some further calculations:

Contribution margin ratio for the original scenario:

$$= \frac{€0.02}{0.50} = 0.04$$

BEP in sales revenue in the original scenario:

$$= \frac{€40\,000}{0.04} = €1\,000\,000 \text{ (check: } 2\,000\,000 \text{ litres @ €0.50)}$$

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Review Questions and Case Studies

We'll now discuss the technique be applied to a business making and selling more than one product

Example

A business manufactures and sells two products which have the following revenue and cost profiles:

	Big		Small
Sales price	€10		€6
Variable costs	8		3
Contribution margin	€2		€3
Total fixed costs		€50 000	
Estimated sales for year	20 000		20 000

Note that Small has a higher contribution margin per unit and the business would prefer to sell more of product Small than product Big. However the sales team are budgeting for equal sales of Big and Small.

We'll now discuss the technique be applied to a business making and selling more than one product

The budgeted profit figure and the break-even point can be calculated easily:

	Big	Small	Total
Sales units	20 000	20 000	40 000
	€	€	€
Sales revenue	200 000	120 000	320 000
Variable costs	160 000	60 000	220 000
Contribution	40 000	60 000	100 000
Fixed cost			50 000
Budgeted profit			50 000
Contribution margin ratio	0.2	0.5	0.3125
Break-even point in sales revenue		$\text{€}50\,000 / 0.3125$ $= \text{€}160\,000$	

Note the weighted average nature of this figure because neither product has a contribution margin ratio of 0.3125; Big's is substantially less and Small's is substantially greater. But the figure of €160,000 represents the break-even point of this business provided the sales of Big and Small match each other unit for unit.

We'll now discuss the technique be applied to a business making and selling more than one product

However this is an unrealistic proposition; the likelihood is that one product will always outpace another for reasons which were unforeseen at the planning stage. Consider how the above numbers change when 25 000 Big units and 15 000 Small units are sold:

	Big	Small	Total
Sales units	25 000	15 000	40 000
	€	€	€
Sales revenue	250 000	90 000	340 000
Variable costs	200 000	45 000	245 000
Contribution	50 000	45 000	95 000
Fixed cost			50 000
Budgeted profit			45 000
Contribution margin ratio	0.2	0.5	0.2794
Break-even point in sales revenue		$\frac{€50\,000}{0.2794}$ $= €178\,955$	

The product switch in the market away from Small to Big has caused the weighted average contribution margin ratio to fall which, in turn, pushes up the break-even point in sales revenue despite the fact that the total number of sales units has remained the same at 40,000.

We'll now discuss the technique be applied to a business making and selling more than one product

A similar calculation can be made using contribution margin per unit as opposed to contribution margin ratio:

	Big	Small	Total
Sales units	20 000	20 000	40 000
	€	€	€
Sales revenue	200 000	120 000	320 000
Variable costs	160 000	60 000	220 000
Contribution	40 000	60 000	100 000
Fixed cost			50 000
Budgeted profit			50 000
Contribution margin	€2	€3	€2.50
Break-even point in units		$\text{€}50\,000 / \text{€}2.50$ $= 20\,000$	

Thus, provided the business sells 10,000 units of Big and 10,000 units of Small, the break-even point will be at the 20 000 level, that is, the budgeted sales mix must be held constant.

What managerial information do these techniques produce when the slightest variation in sales mix will throw out the budgeted profit and break-even point?

The process of fact finding and planning lying behind the calculation

→
*More important
than*

Absolute numbers (units or revenue) which is the BEP of €160,000

The real advantage

Awareness of what sales mix caused a change in bottom-line profit or break-even point

How best to capitalize on the revenue-generating potential of individual products

For example, management in the business used in the example should endeavor to maximize sales of Small at the expense of Big (because Small earns a greater contribution than Big) even if this may drive up fixed costs in total. In other words, managers must use accounting numbers as signals to direct their managerial efforts and not allow them to act as blinkers which narrow managerial focus and action.

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The management of the business want to concentrate its efforts to maximize contribution

Logical Conclusion

Product Small because of its superior contribution in earning capacity (earns €1 more contribution than Big)

Argument

This conclusion is simplistic because it ignores the numerous ***constraints or factors*** which limit operations



The availability of space (in a supermarket) or machine hours (car assembly) or skilled labor (siliconwafer fabrication)

Limiting factor

The ultimate bottleneck through which production flows and which constrains management from producing unlimited quantities even though the market demand is buoyant

The secret of good management is to identify the limiting factor and calculate contribution margin per unit of limiting factor

The following example shows how to identify the limiting factor and calculate contribution margin per unit of limiting factor

Example

Consider again the business producing products Big and Small:

	Big	Small
Sales revenue	€10	€6
Variable cost	8	3
Contribution	<u>€2</u>	<u>€3</u>

The management of this business would be well advised on the basis of this information alone to focus all manufacturing and selling effort on Small because of its superior contribution margin per unit. However if the limiting factor of this business is machine hours, the picture changes:

	Big	Small
Contribution margin per unit	€2	€3
Machine hours per unit	1	2
Total machine hours available 60 000		
Contribution margin per limiting factor	€2	€1.50

The following example shows how to identify the limiting factor and calculate contribution margin per unit of limiting factor

Example (Continued)

At the extreme, and ignoring market demand for either product, the company could produce and sell either:

Big: $60\,000 \text{ units} \times 1 \text{ hour per unit} \times \text{€}2 \text{ contribution per hour}$
 $= \text{€}120\,000 \text{ contribution}$

Small: $30\,000 \text{ units} \times 2 \text{ hours per unit} \times \text{€}1.50 \text{ contribution per hour}$
 $= \text{€}90\,000 \text{ contribution}$

Subject to the demands of the market place, the management of this business should concentrate on product Big because of its superior contribution margin per limiting factor of production. The objective is to maximize contribution per unit of limiting factor.

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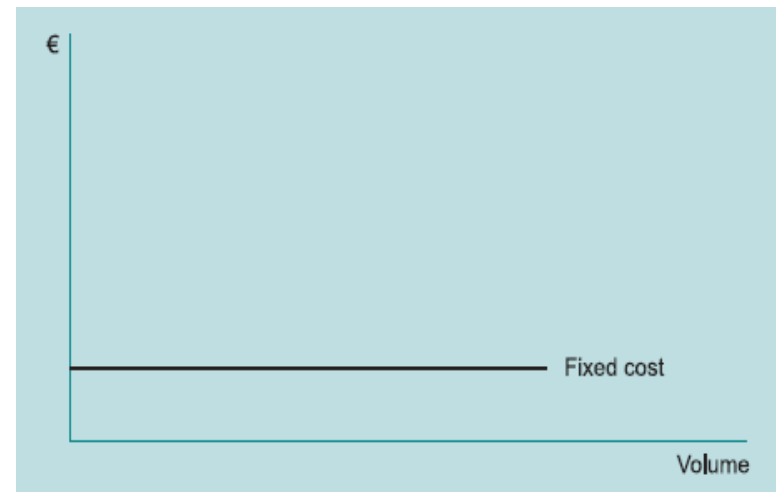
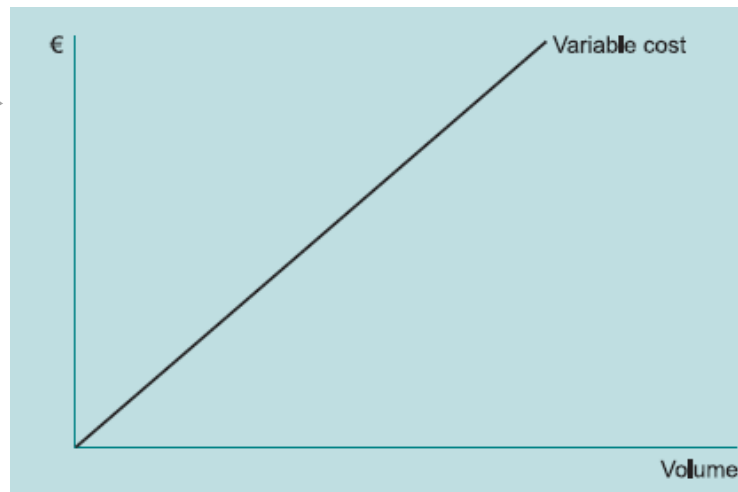
Now let's review the underlying assumptions implicit in our analysis of the break-even chart

Assumptions

1 All costs can be identified as variable or fixed

A greater proportion of costs assume the character of fixed costs, particularly as the time horizon shortens; managerial action is not so flexible as the term 'variable cost' implies

2 All costs behave as depicted in the following charts along the whole cost curve

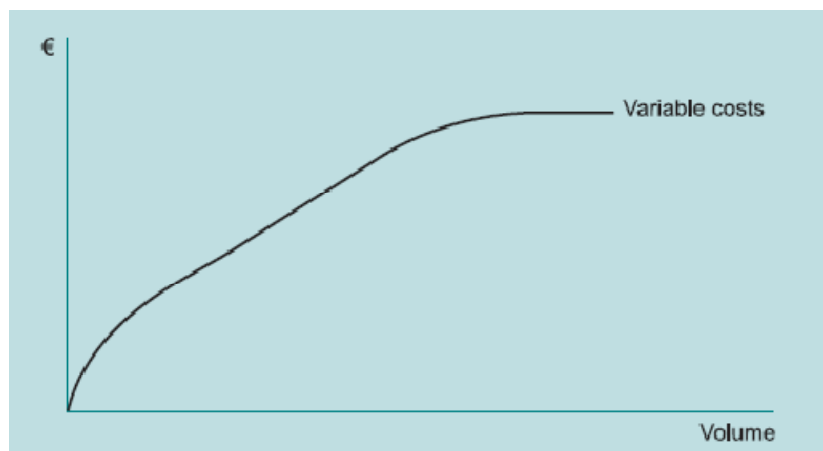


In reality the costs do not behave as smoothly as the charts imply

Variable Costs

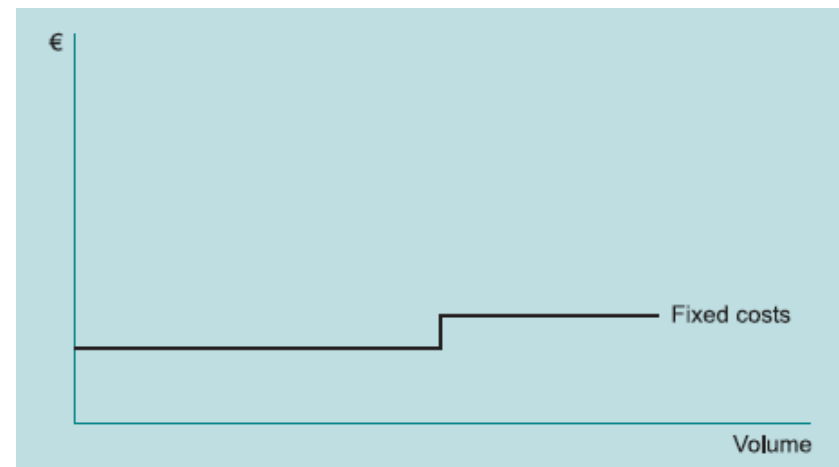
The beginning of the production cycle:
Steeper because machines would not be calibrated correctly and workers would be tentative and unsure of their roles

As the volume is pushed up towards capacity, the production is working smoothly and the company could be benefiting from volume discounts in raw materials

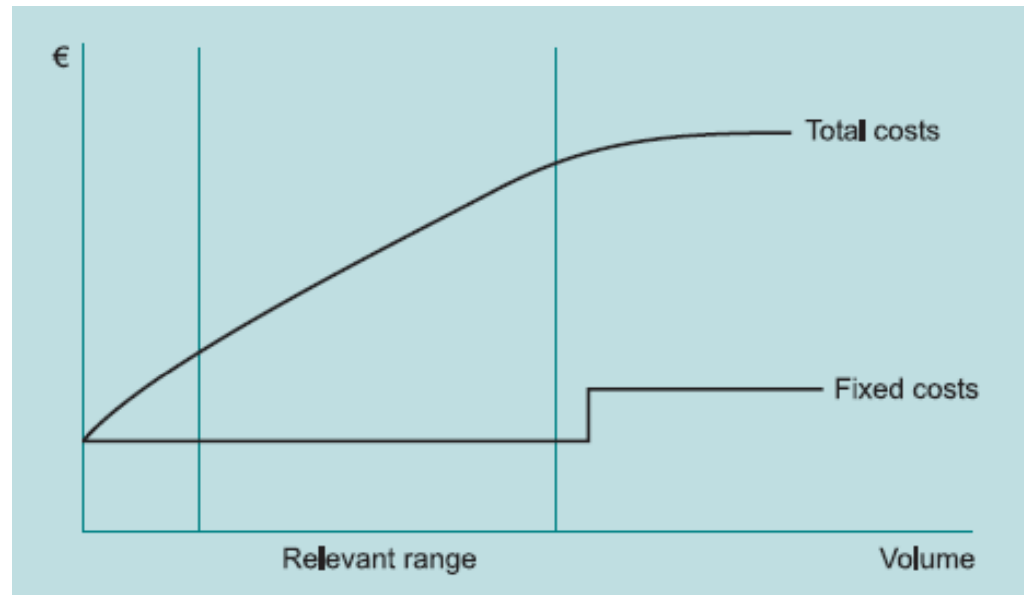


Fixed Costs

The curve would take a step up as maximum capacity is approached: another machine would be required, additional maintenance would be necessary; space would have to be increased



Putting these diagrams together reveals that there is a range in the middle where the fixed and variable costs behave as their defining adjectives describe



We call this the relevant range within which our assumptions and conclusions are valid

Now let's review the underlying assumptions implicit in our analysis of the break-even chart

Assumptions

3 Sales price per unit remains unchanged.

It would be anticipated that a company would alter its sales price per unit to reflect changing market conditions, volume orders and long-term customer commitment

4 Sales mix will be maintained precisely as budgeted

5 All production is sold

This assumption is required so that all costs are moved to the market place and not retained in inventory. (Remember: profit cannot be earned on unsold inventory.)

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